MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER SCHOTTKY

HEX BUS DRIVERS WITH 3-STATE OUTPUTS,

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power schottky, hex bus driver (3 state) microcircuits. Three product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Hex bus driver, gated enable inputs for x-y coincident bus control
02	Hex inverter bus driver, gated enable inputs for x-y coincident bus control
03	Hex bus driver, 4-line and 2-line enable inputs
04	Hex inverter bus driver, 4-line and 2-line enable inputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead 1/4" x 7/8" dual-in-line package)
F	F-5 (16-lead 1/4" x 3/8" flat package)

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to 7.0 V
Input voltage range	-1.5 V at -18 mA to 5.5 V
Storage temperature range	-65°C to 150°C
Maximum power dissipation (P_D) 1/	
Device types 01 and 03	132 mW
Device types 02 and 04	116 mW

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case θ_{JC}:	
Case E	0.08°C/mW
Case F	0.09°C/mW
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Normalized fanout (each input) 2/	20 maximum
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

2/ The device will fan out in both high and low levels to the specified number of inputs for the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IN} = 0.7 V or 2.0 V; I_{OH} = -1 mA	All	2.4	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IN} = 2.0 V or 0.7 V; I_{OL} = 12 mA	All	---	0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA; T_C = 25°C	All	---	-1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V	All	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	All	---	0.1	mA
Inhibited-state output leakage current	I_{O1}	V_{CC} = 5.5 V; V_{IN} = 2.0 V; V_{OUT} = 2.4 V	All	---	20	µA
	I_{O2}	V_{CC} = 5.5 V; V_{IN} = 2.0 V; V_{OUT} = 0.4 V	All	---	-20	µA
Low-level input current at A input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V; $\overline{E}$ inputs at 0.4 V	All	-30	-400	µA
Low-level input current at $\overline{E}$ input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	All	-1.0	-400	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	All	-15	-225	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01	---	24	mA
			02	---	21	mA
			03	---	24	mA
			04	---	21	mA
Propagation delay time (low-to-high level output)	t_{PLH}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	24	ns
			02	2	22	ns
			03	2	24	ns
			04	2	22	ns
Propagation delay time (high-to-low level output)	t_{PHL}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	31	ns
			02	2	26	ns
			03	2	31	ns
			04	2	26	ns
Propagation delay time (disabled to high-level output)	t_{PZH}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	45	ns
			02	2	45	ns
			03	2	45	ns
			04	2	45	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
Propagation delay time (high-level to disabled output)	t_{PHZ}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	39	ns
			02	2	42	ns
			03	2	39	ns
			04	2	42	ns
Propagation delay time (disabled to low-level output)	t_{PZL}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	52	ns
			02	2	58	ns
			03	2	52	ns
			04	2	58	ns
Propagation delay time (low-level to disabled output)	t_{PLZ}	V_{CC} = 5.0 V, R_L = See fig. 4 C_L = 50 pF	01	2	45	ns
			02	2	45	ns
			03	2	45	ns
			04	2	45	ns

1/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening and quality conformance inspection by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical tests (pre burn-in) (method 5004)	1	1	None
Final electrical tests (method 5004)	1*,2,3, 9,10,11	1*,2,3, 9	1
Group A electrical tests (method 5005)	1,2,3, 9,10,11	1,2,3, 9	1,2,3, 9
Group B electrical tests (method 5005) (subgroup 5)	1,2,3, 9,10,11	N/A	N/A
Group C end-point electrical tests (method 5005)	N/A	1,2,3	1
Additional electrical tests for group C inspections	N/A	10,11	10,11
Group D end-point electrical tests (method 5005)	1,2,3	1,2,3	1

* PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D or E, using the circuit shown on figure 3, or equivalent.

2. T_A = 125°C minimum.

b. Interim and final electrical tests shall be as specified in table II; interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical tests shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

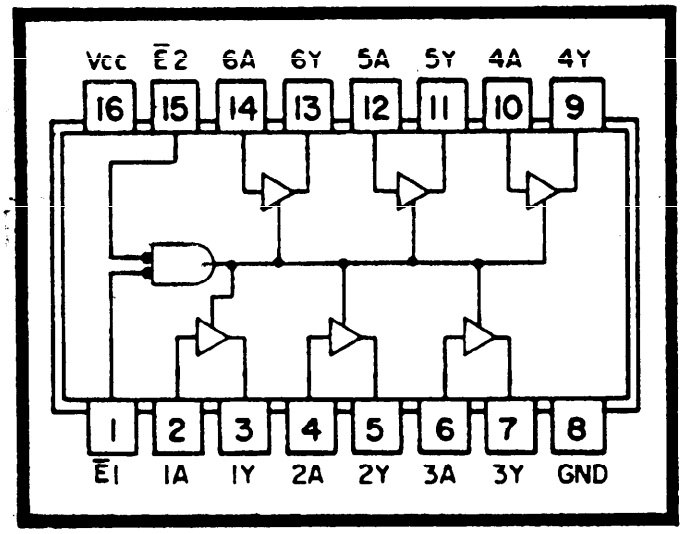
(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = 125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

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Device type 01



Device type 02

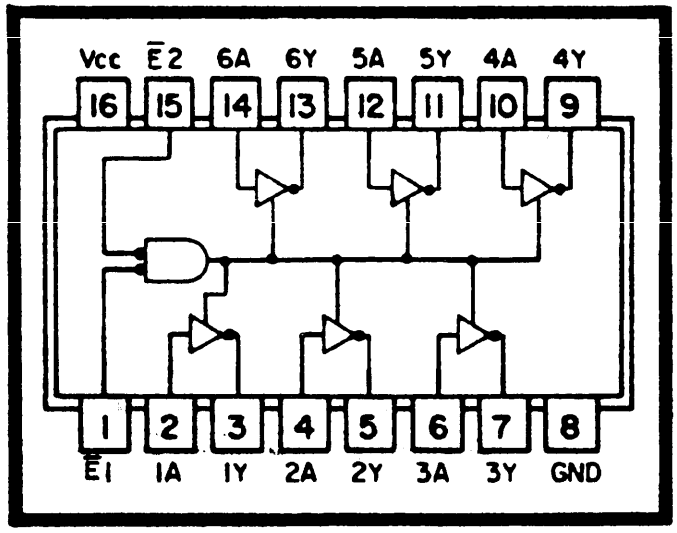
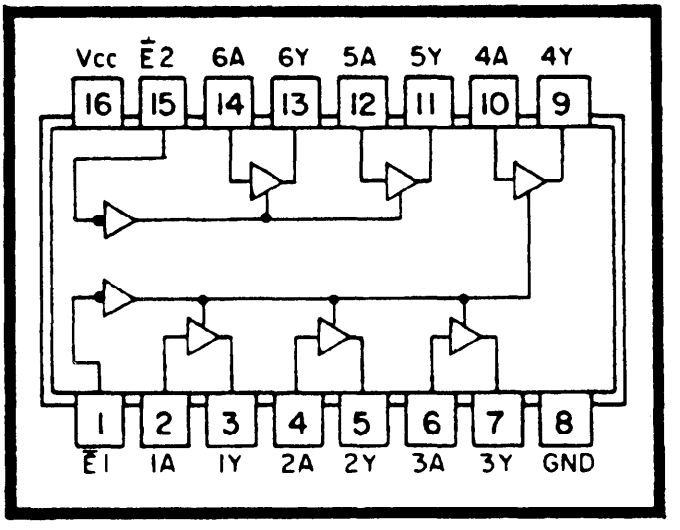


FIGURE 1. Terminal connections and logic diagram - top view.

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Device type 03



Device type 04

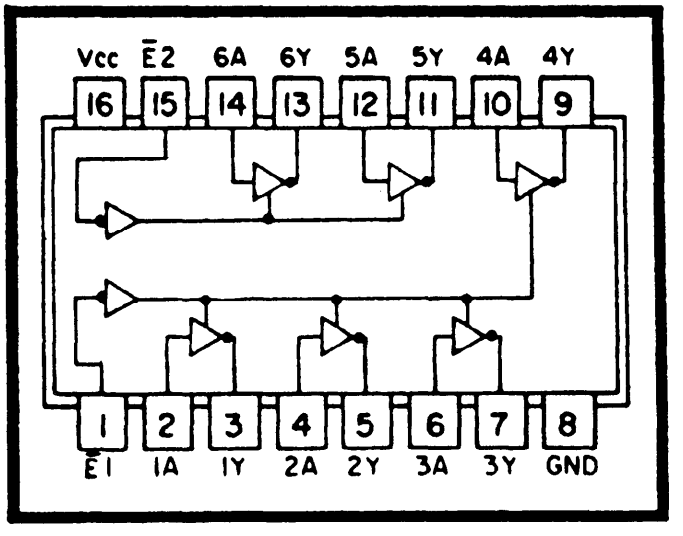


FIGURE 1. Terminal connections and logic diagram - top view - Continued.

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Device type 01

INPUTS			OUTPUT
$\overline{E}1$	$\overline{E}2$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	H
L	L	L	L

Device type 02

INPUTS			OUTPUT
$\overline{E}1$	$\overline{E}2$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	L
L	L	L	H

Device type 03

INPUTS		OUTPUT
$\overline{E}$	A	Y
H	X	Z
L	H	H
L	L	L

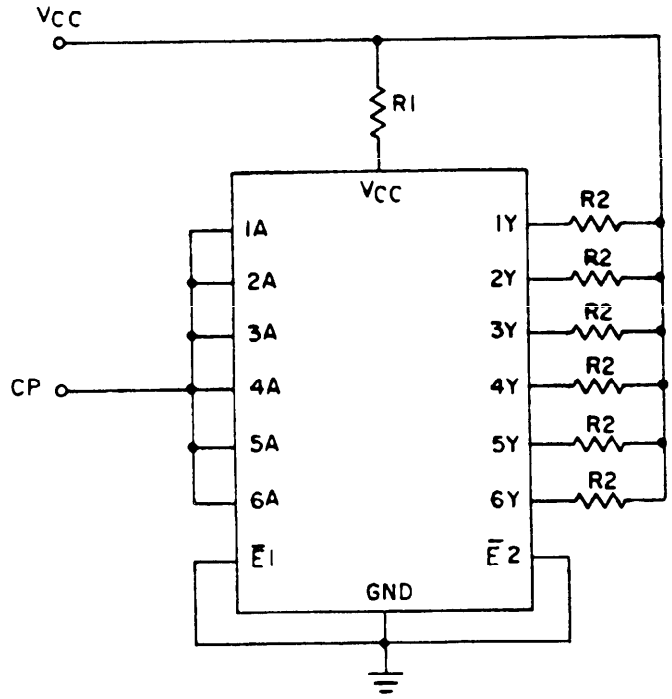
Device type 04

INPUTS		OUTPUT
$\overline{E}$	A	Y
H	X	Z
L	H	L
L	L	H

H = High level
L = Low level
X = Irrelevant
Z = High-impedance

FIGURE 2. Truth tables (each driver).

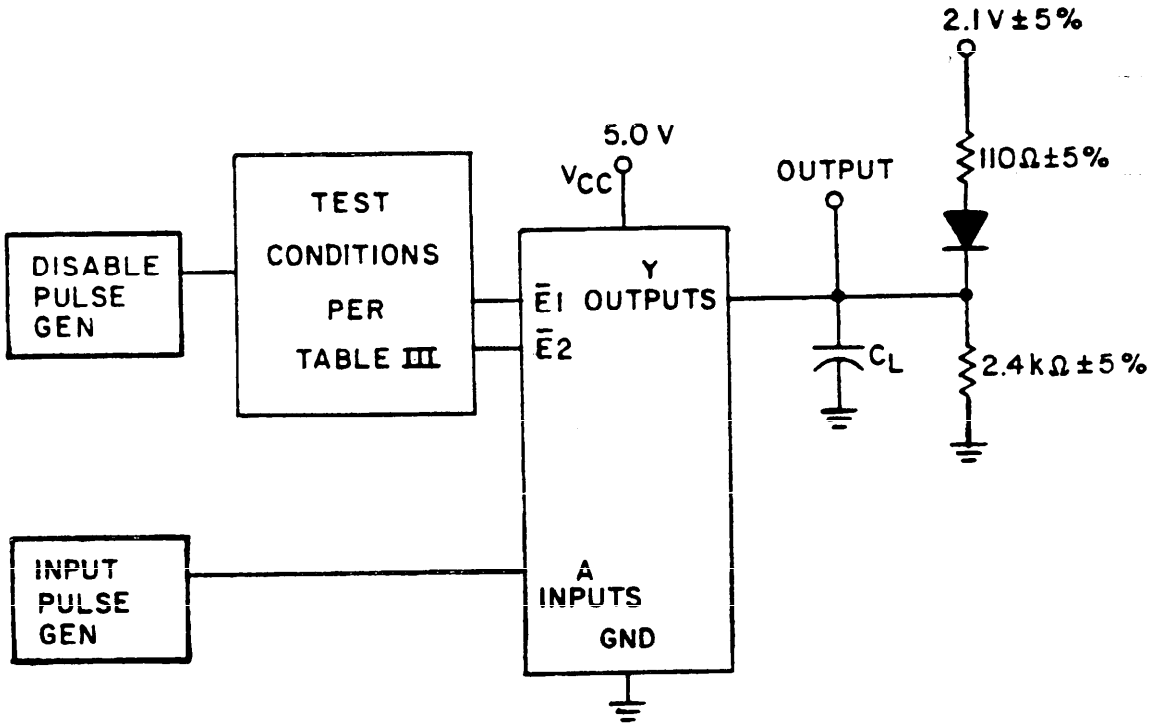
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NOTES:
1. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 3. Burn-in and life test circuit.

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NOTES:
1. C_L = 50 pF ±10%, including jig and probe capacitance.
2. All diodes are 1N3064 or equivalent.

FIGURE 4. Switching time test circuit and waveforms.

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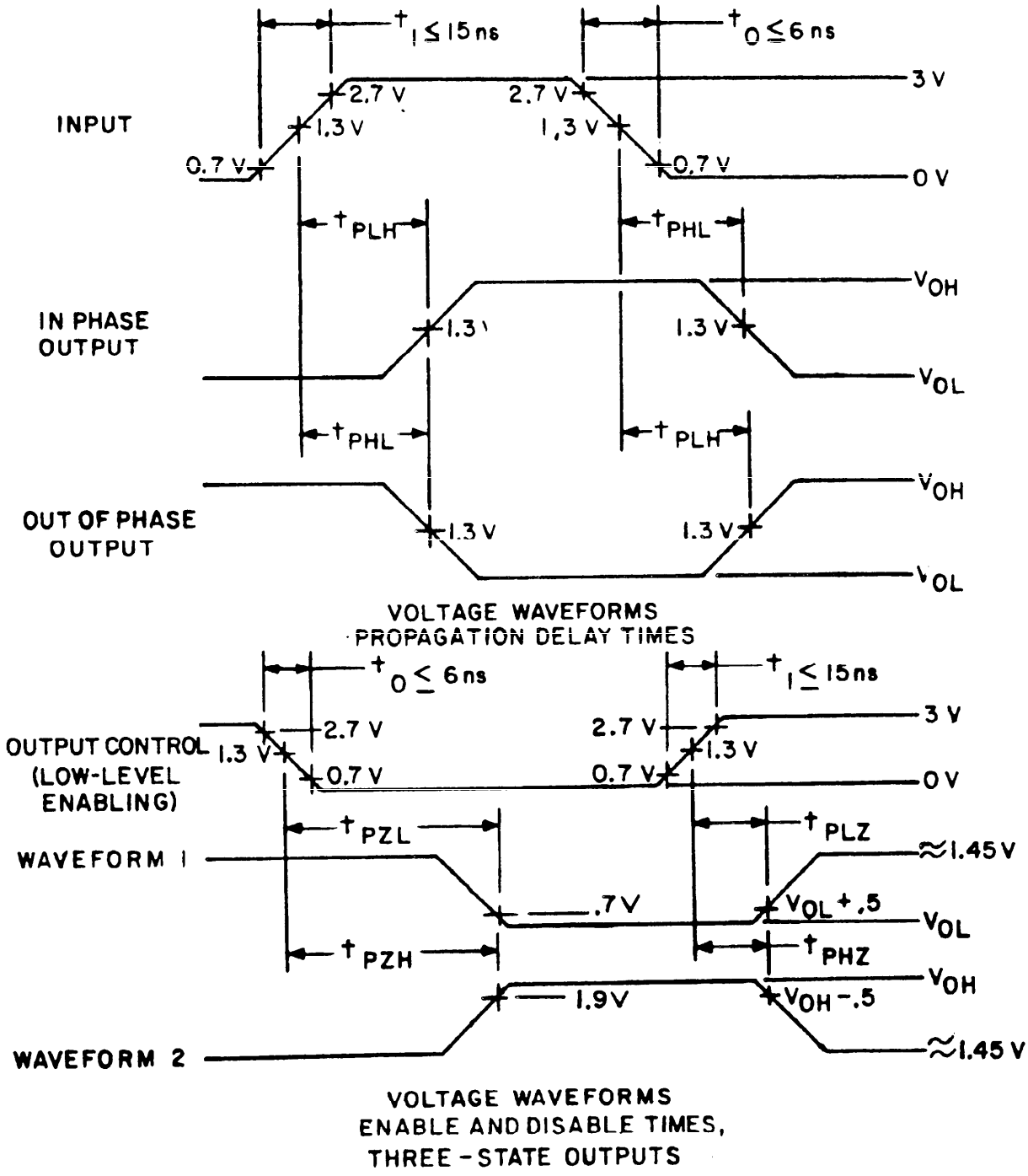


FIGURE 4. Switching time test circuit and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	0.7 V	2.0 V	-1 mA	2.0 V	-1 mA	2.0 V	-1 mA	GND	-1 mA	2.0 V	-1 mA	2.0 V	-1 mA	2.0 V	0.7 V	4.5 V	1Y	2.4		V dc
			2	"														"	"	2Y	"		
			3	"														"	"	3Y	"		
			4	"														"	"	4Y	"		
			5	"														"	"	5Y	"		
			6	"														"	"	6Y	"		
	V_{OL}	3007	7	"	0.7 V	12 mA	0.7 V	12 mA	0.7 V	12 mA	"	12 mA	0.7 V	12 mA	0.7 V	12 mA	0.7 V	"	"	1Y		0.4	
			8	"														"	"	2Y		"	
			9	"														"	"	3Y		"	
			10	"														"	"	4Y		"	
			11	"														"	"	5Y		"	
			12	"														"	"	6Y		"	
	I_{O1}		13	2.0 V	2.0 V	2.4 V												0.7 V	5.5 V	1Y		20	μA
			14	0.7 V	0.7 V	2.4 V												2.0 V	"	1Y		"	
			15	2.0 V			2.0 V	2.4 V										0.7 V	"	2Y		"	
			16	0.7 V			0.7 V	2.4 V										2.0 V	"	2Y		"	
			17	2.0 V					2.0 V	2.4 V								0.7 V	"	3Y		"	
			18	0.7 V					0.7 V	2.4 V								2.0 V	"	3Y		"	
			19	2.0 V								2.4 V	2.0 V					0.7 V	"	4Y		"	
			20	0.7 V								2.4 V	0.7 V					2.0 V	"	4Y		"	
			21	2.0 V										2.4 V	2.0 V			0.7 V	"	5Y		"	
			22	0.7 V										2.4 V	0.7 V			2.0 V	"	5Y		"	
			23	2.0 V												2.4 V	2.0 V	0.7 V	"	6Y		"	
			24	0.7 V												2.4 V	0.7 V	2.0 V	"	6Y		"	
	I_{O2}		25	2.0 V	0.7 V	0.4 V												0.7 V	"	1Y		-20	
			26	0.7 V	2.0 V	0.4 V												2.0 V	"	1Y		"	
			27	2.0 V			0.7 V	0.4 V										0.7 V	"	2Y		"	
			28	0.7 V			2.0 V	0.4 V										2.0 V	"	2Y		"	
			29	2.0 V					0.7 V	0.4 V								0.7 V	"	3Y		"	
			30	0.7 V					2.0 V	0.4 V								2.0 V	"	3Y		"	
			31	2.0 V								0.4 V	0.7 V					0.7 V	"	4Y		"	
			32	0.7 V								0.4 V	2.0 V					2.0 V	"	4Y		"	
			33	2.0 V										0.4 V	0.7 V			0.7 V	"	5Y		"	
			34	0.7 V										0.4 V	2.0 V			2.0 V	"	5Y		"	
			35	2.0 V												0.4 V	0.7 V	0.7 V	"	6Y		"	
			36	0.7 V												0.4 V	2.0 V	2.0 V	"	6Y		"	
	V_{IC}		37		-18 mA		-18 mA		-18 mA										4.6 V	1A		-1.5	V dc
			38																"	2A		"	
			39																"	3A		"	
			40										-18 mA						"	4A		"	
			41																"	5A		"	
			42												-18 mA				"	6A		"	
			43	-18 mA													-18 mA		"	$\overline{E1}$		"	
			44															-18 mA	"	$\overline{E2}$		"	
	I_{IH1}	3010	45	2.7 V															5.5 V	$\overline{E1}$		20	μA
			46		2.7 V														"	1A		"	
			47				2.7 V												"	2A		"	
			48						2.7 V										"	3A		"	
			49										2.7 V						"	4A		"	
			50												2.7 V				"	5A		"	
			51														2.7 V		"	6A		"	
			52															2.7 V	"	$\overline{E2}$		"	
	I_{IH2}		53	5.5 V															"	$\overline{E1}$		0.1	mA
			54		5.5 V														"	1A		"	
			55				5.5 V												"	2A		"	
			56						5.5 V										"	3A		"	
			57										5.5 V						"	4A		"	

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}				
1 T_C = 25°C	I_{IH2}	3010	58								GND				5.5 V		5.5 V		5.5 V	5A		0.1	mA
	↓	"	59								"				"		"	5.5 V	"	6A		"	"
		"	60								"				"		"	"	"	$\overline{E}2$		"	"
	I_{IL1}	3009	61	0.4 V	0.4 V		0.4 V		0.4 V		"		0.4 V		0.4 V		0.4 V	0.4 V	"	1A	1/	1/	µA
	↓	"	62	"							"							"	"	2A	"	"	"
		"	63	"							"							"	"	3A	"	"	"
		"	64	"							"							"	"	4A	"	"	"
		"	65	"							"							"	"	5A	"	"	"
		"	66	"							"							"	"	6A	"	"	"
	I_{IL2}	"	67	"							"							0.4 V	"	$\overline{E}1$	"	"	"
	↓	"	68	"							"							"	"	$\overline{E}2$	"	"	"
	I_{OS}	3011	69	0.7 V	5.5 V	GND	5.5 V	GND	5.5 V	GND	"	GND	5.5 V	GND	5.5 V	GND	5.5 V	0.7 V	"	1Y	2/	2/	mA
	↓	"	70	"							"							"	"	2Y	"	"	"
		"	71	"							"							"	"	3Y	"	"	"
		"	72	"							"							"	"	4Y	"	"	"
		"	73	"							"							"	"	5Y	"	"	"
		"	74	"							"							"	"	6Y	"	"	"
	I_{CC}	3005	75	4.5 V	GND		GND		GND		"		GND		GND		GND	4.5 V	"	V_{CC}		24	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = 125°C.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH}	3003 (Fig. 4)	76	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	16	ns
	↓	"	77	"							"							"	"	2A - 2Y	"	"	"
		"	78	"							"							"	"	3A - 3Y	"	"	"
		"	79	"							"							"	"	4A - 4Y	"	"	"
		"	80	"							"							"	"	5A - 5Y	"	"	"
		"	81	"							"							"	"	6A - 6Y	"	"	"
	t_{PHL}	"	82	"	3/	OUT	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	"	"	1A - 1Y	"	22	"
	↓	"	83	"							"							"	"	2A - 2Y	"	"	"
		"	84	"							"							"	"	3A - 3Y	"	"	"
		"	85	"							"							"	"	4A - 4Y	"	"	"
		"	86	"							"							"	"	5A - 5Y	"	"	"
		"	87	"							"							"	"	6A - 6Y	"	"	"
	t_{PZH}	"	88	4/	4.5 V	OUT					"							"	"	$\overline{E}1$ - 1Y	"	35	"
	t_{PZL}	"	89	GND	GND	"					"							4/	"	$\overline{E}2$ - 1Y	"	40	"
	t_{PHZ}	"	90	4/	4.5 V	"					"							GND	"	$\overline{E}1$ - 1Y	"	30	"
	t_{PLZ}	"	91	4/	GND	"					"							"	"	$\overline{E}1$ - 1Y	"	35	"
10 T_C = 125°C	t_{PLH}	"	92	GND	3/		3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	"	"	1A - 1Y	"	21	"
	↓	"	93	"							"							"	"	2A - 2Y	"	"	"
		"	94	"							"							"	"	3A - 3Y	"	"	"
		"	95	"							"							"	"	4A - 4Y	"	"	"
		"	96	"							"							"	"	5A - 5Y	"	"	"
		"	97	"							"							"	"	6A - 6Y	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				E1	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	E2	V_{CC}				
10 T_C = 125°C	t_{PHL}	3003 (Fig. 4)	98	GND	2/	OUT	2/	OUT	2/	OUT	GND	OUT	2/	OUT	2/	OUT	2/	GND	5.0 V	1A - 1Y	2	29	ns
		"	99	"							"							"	"	2A - 2Y	"	"	"
		"	100	"							"							"	"	3A - 3Y	"	"	"
		"	101	"							"							"	"	4A - 4Y	"	"	"
		"	102	"							"							"	"	5A - 5Y	"	"	"
		"	103	"							"							"	"	6A - 6Y	"	"	"
	t_{PZH}	"	104	3/	4.5 V	OUT					"							"	"	E1 - 1Y	"	45	"
	t_{PZL}	"	105	GND	GND	"					"							3/	"	E2 - 1Y	"	52	"
	t_{PHZ}	"	106	3/	4.5 V	"					"							GND	"	E1 - 1Y	"	39	"
	t_{PLZ}	"	107	3/	GND	"					"							GND	"	E1 - 1Y	"	45	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ I_{IL} limits (μA) min/max values for circuits are shown below.
2/ I_{OS} limits (mA) min/max values for circuits are shown below.

Parameter	Terminals	Circuits A	B	C	D	E	F
I_{IL1}	1A,2A,3A,4A,5A,6A	-150/-380	-30/-300	-160/-400	-160/-400	-160/-400	-160/-400
I_{IL2}	E1, E2	-1.0/-150	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{OS}	1Y,2Y,3Y,4Y,5Y,6Y	-40/-225	-30/-130	-30/-130	-15/-100	-30/-130	-15/-100

3/ Pulse from input pulse generator.
4/ Pulse from disable generator.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				E1	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	E2	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	0.7 V	0.7 V	-1 mA	0.7 V	-1 mA			GND							0.7 V	4.5 V	1Y	2.4		V dc
		"	2	"					0.7 V	-1 mA	"							"	"	2Y	"		"
		"	3	"							"	-1 mA	0.7 V					"	"	3Y	"		"
		"	4	"							"			-1 mA	0.7 V			"	"	4Y	"		"
		"	5	"							"					-1 mA	0.7 V	"	"	5Y	"		"
		"	6	"							"							"	"	6Y	"		"
	V_{OL}	3007	7	"	2.0 V	12 mA					"							"	"	1Y		0.4	"
		"	8	"			2.0 V	12 mA			"							"	"	2Y		"	"
		"	9	"					2.0 V	12 mA	"							"	"	3Y		"	"
		"	10	"							"	12 mA	2.0 V					"	"	4Y		"	"
		"	11	"							"			12 mA	2.0 V			"	"	5Y		"	"
		"	12	"							"					12 mA	2.0 V	"	"	6Y		"	"
	I_{O1}		13	2.0 V	2.0 V	2.4 V												0.7 V	5.5 V	1Y		20	µA
			14	0.7 V	0.7 V	2.4 V												2.0 V	"	1Y		"	"
			15	2.0 V			2.0 V	2.4 V										0.7 V	"	2Y		"	"
			16	0.7 V			0.7 V	2.4 V										2.0 V	"	2Y		"	"
			17	2.0 V					2.0 V	2.4 V								0.7 V	"	3Y		"	"
			18	0.7 V					0.7 V	2.4 V								2.0 V	"	3Y		"	"
			19	2.0 V								2.4 V	2.0 V					0.7 V	"	4Y		"	"
			20	0.7 V								2.4 V	0.7 V					2.0 V	"	4Y		"	"
			21	2.0 V										2.4 V	2.0 V			0.7 V	"	5Y		"	"
			22	0.7 V										2.4 V	0.7 V			2.0 V	"	5Y		"	"
			23	2.0 V												2.4 V	2.0 V	0.7 V	"	6Y		"	"
			24	0.7 V												2.4 V	0.7 V	2.0 V	"	6Y		"	"
	I_{O2}		25	2.0 V	0.7 V	0.4 V												0.7 V	"	1Y		-20	"
			26	0.7 V	2.0 V	0.4 V												2.0 V	"	1Y		"	"
			27	2.0 V			0.7 V	0.4 V										0.7 V	"	2Y		"	"
			28	0.7 V			2.0 V	0.4 V										2.0 V	"	2Y		"	"
			29	2.0 V					0.7 V	0.4 V								0.7 V	"	3Y		"	"
			30	0.7 V					2.0 V	0.4 V								2.0 V	"	3Y		"	"
			31	2.0 V								0.4 V	0.7 V					0.7 V	"	4Y		"	"
			32	0.7 V								0.4 V	2.0 V					2.0 V	"	4Y		"	"
			33	2.0 V										0.4 V	0.7 V			0.7 V	"	5Y		"	"
			34	0.7 V										0.4 V	2.0 V			2.0 V	"	5Y		"	"
			35	2.0 V												0.4 V	0.7 V	0.7 V	"	6Y		"	"
			36	0.7 V												0.4 V	2.0 V	2.0 V	"	6Y		"	"
	V_{IC}		37		-18 mA														4.5 V	1A		-1.5	V dc
			38				-18 mA												"	2A		"	"
			39						-18 mA										"	3A		"	"
			40																"	4A		"	"
			41										-18 mA						"	5A		"	"
			42												-18 mA				"	6A		"	"
			43	-18 mA													-18 mA		"	E1		"	"
			44															-18 mA	"	E2		"	"
	I_{IH1}	3010	45	2.7 V															5.5 V	E1		20	µA
		"	46		2.7 V														"	1A		"	"
		"	47				2.7 V												"	2A		"	"
		"	48						2.7 V										"	3A		"	"
		"	49										2.7 V						"	4A		"	"
		"	50												2.7 V				"	5A		"	"
		"	51														2.7 V		"	6A		"	"
		"	52															2.7 V	"	E2		"	"
	I_{IH2}	"	53	5.5 V															"	E1		0.1	mA
		"	54		5.5 V														"	1A		"	"
		"	55				5.5 V												"	2A		"	"
		"	56						5.5 V										"	3A		"	"
		"	57										5.5 V						"	4A		"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}				
1 T_C = 25°C	I_{IH2}	3011	58								GND				5.5 V		5.5 V		5.5 V	5A		0.1	mA
	↓	"	59								"							5.5 V	"	6A		"	"
		"	60								"									$\overline{E}2$		"	"
	I_{IL1}	3009	61	0.4 V	0.4 V		0.4 V		0.4 V		"		0.4 V		0.4 V		0.4 V	0.4 V	"	1A	1/	1/	μA
	↓	"	62	"							"							"	"	2A	"	"	"
		"	63	"							"							"	"	3A	"	"	"
		"	64	"							"							"	"	4A	"	"	"
		"	65	"							"				0.4 V			"	"	5A	"	"	"
		"	66	"							"							"	"	6A	"	"	"
	I_{IL2}	"	67	"							"							0.4 V	"	$\overline{E}1$	"	"	"
	↓	"	68	"							"							"	"	$\overline{E}2$	"	"	"
	I_{OS}	3011	69	0.7 V	GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	0.7 V	"	1Y	2/	2/	mA
	↓	"	70	"							"							"	"	2Y	"	"	"
		"	71	"							"							"	"	3Y	"	"	"
		"	72	"							"							"	"	4Y	"	"	"
		"	73	"							"							"	"	5Y	"	"	"
		"	74	"							"							"	"	6Y	"	"	"
	I_{CC}	3005	75	4.5 V	GND		GND		GND		"		GND		GND		GND	4.5 V	"	V_{CC}		21	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = 125°C.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH}	3003 (Fig. 4)	76	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	15	ns
	↓	"	77	"							"							"	"	2A - 2Y	"	"	"
		"	78	"							"							"	"	3A - 3Y	"	"	"
		"	79	"							"							"	"	4A - 4Y	"	"	"
		"	80	"							"							"	"	5A - 5Y	"	"	"
		"	81	"							"							"	"	6A - 6Y	"	"	"
	t_{PHL}	"	82	"	3/	OUT	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	"	"	1A - 1Y	"	18	"
	↓	"	83	"							"							"	"	2A - 2Y	"	"	"
		"	84	"							"							"	"	3A - 3Y	"	"	"
		"	85	"							"							"	"	4A - 4Y	"	"	"
		"	86	"							"							"	"	5A - 5Y	"	"	"
		"	87	"							"							"	"	6A - 6Y	"	"	"
	t_{PZH}	"	88	4/	GND	OUT					"							"	"	$\overline{E}1$ - 1Y	"	35	"
	t_{PZL}	"	89	GND	4.5 V	"					"							4/	"	$\overline{E}2$ - 1Y	"	45	"
	t_{PHZ}	"	90	4/	GND	"					"							GND	"	$\overline{E}1$ - 1Y	"	32	"
	t_{PLZ}	"	91	4/	4.5 V	"					"							"	"	$\overline{E}1$ - 1Y	"	35	"
10 T_C = 125°C	t_{PLH}	"	92	GND	3/	"	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	"	"	1A - 1Y	"	20	"
↓	↓	"	93	"							"							"	"	2A - 2Y	"	"	"
		"	94	"							"							"	"	3A - 3Y	"	"	"
		"	95	"							"							"	"	4A - 4Y	"	"	"
		"	96	"							"							"	"	5A - 5Y	"	"	"
		"	97	"							"							"	"	6A - 6Y	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}				
10 T_C = 125°C	t_{PHL}	3003 (Fig. 4	98	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	23	ns
			99	"							"							"	"	2A - 2Y	"	"	"
			100	"							"							"	"	3A - 3Y	"	"	"
			101	"							"							"	"	4A - 4Y	"	"	"
			102	"							"							"	"	5A - 5Y	"	"	"
			103	"							"							"	"	6A - 6Y	"	"	"
	t_{PZH}	"	104	4/	GND	OUT					"							"	"	$\overline{E}1$ - 1Y	"	45	"
	t_{PZL}	"	105	GND	4.5 V	"					"							4/	"	$\overline{E}2$ - 1Y	"	58	"
	t_{PHZ}	"	106	4/	GND	"					"							GND	"	$\overline{E}1$ - 1Y	"	42	"
	t_{PLZ}	"	107	4/	4.5 V	"					"							GND	"	$\overline{E}1$ - 1Y	"	45	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ I_{IL} limits (μA) min/max values for circuits are shown below.
2/ I_{OS} limits (mA) min/max values for circuits are shown below.

Parameter	Terminals	Circuits A	B	C	D	E	F
I_{IL1}	1A,2A,3A,4A,5A,6A	-150/-380	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{IL2}	$\overline{E}1$, $\overline{E}2$	-1.0/-150	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{OS}	1Y,2Y,3Y,4Y,5Y,6Y	-40/-225	-30/-130	-30/-130	-15/-100	-30/-130	-15/-100

3/ Pulse from input generator.
4/ Pulse from disable generator.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥ 2.0 V; $I_{OH} \leq 0.7$ V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 E1	2 1A	3 1Y	4 2A	5 2Y	6 3A	7 3Y	8 GND	9 1Y	10 4A	11 5Y	12 5A	13 6Y	14 6A	15 E2	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 $T_C = 25°C$	V_{OH}	3006	1	0.7 V	2.0 V	-1 mA	2.0 V	-1 mA	2.0 V	-1 mA	GND	-1 mA	2.0 V	-1 mA	2.0 V	-1 mA	2.0 V	0.7 V	4.5 V	1Y	2.4		V dc
			2	"							"							0.7 V	"	2Y	"		
			3	"							"								"	3Y	"		
			4	"							"								"	4Y	"		
			5	"							"								"	5Y	"		
			6	"							"								"	6Y	"		
	V_{OL}	3007	7	0.7 V	0.7 V	12 mA	0.7 V	12 mA	0.7 V	12 mA	"	12 mA	0.7 V	12 mA	0.7 V	12 mA	0.7 V	0.7 V	"	1Y		0.4	"
			8	"							"								"	2Y			
			9	"							"								"	3Y			
			10	"							"								"	4Y			
			11	"							"								"	5Y			
			12	"							"								"	6Y			
	I_{O1}		13	2.0 V	2.0 V	2.4 V	2.0 V	2.4 V	2.0 V	2.4 V	"	2.4 V	2.0 V	2.4 V	0.7 V	0.4 V	0.7 V	2.0 V	5.5 V	1Y		20	µA
			14	"	0.7 V		0.7 V		0.7 V		"		0.7 V		2.0 V		2.0 V		"	1Y		"	
			15	"							"								"	2Y		"	
			16	"							"								"	2Y		"	
			17	"							"								"	3Y		"	
			18	"							"								"	3Y		"	
			19	"							"								"	4Y		"	
			20	"							"								"	4Y		"	
			21	"							"								"	5Y		"	
			22	"							"								"	5Y		"	
			23	"							"								"	6Y		"	
			24	"							"								"	6Y		"	
	I_{O2}		25	2.0 V	0.7 V	0.4 V	0.7 V	0.4 V	0.7 V	0.4 V	"	0.4 V	0.7 V	0.4 V	0.7 V	0.4 V	0.7 V	2.0 V	"	1Y		-20	"
			26	"	2.0 V		2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V		"	1Y		"	
			27	"							"								"	2Y		"	
			28	"							"								"	2Y		"	
			29	"							"								"	3Y		"	
			30	"							"								"	3Y		"	
			31	"							"								"	4Y		"	
			32	"							"								"	4Y		"	
			33	"							"								"	5Y		"	
			34	"							"								"	5Y		"	
			35	"							"								"	6Y		"	
			36	"							"								"	6Y		"	
	V_{IC}		37		-18 mA		-18 mA		-18 mA				-18 mA		-18 mA		-18 mA		4.5 V	1A	-1.5	V dc
			38																"	2A		"	
			39																"	3A		"	
			40																"	4A		"	
			41																"	5A		"	
			42																"	6A		"	
			43																"	E1		"	
			44		-18 mA														"	E2		"	
	I_{IH1}	3010	45	2.7 V	2.7 V		2.7 V		2.7 V				2.7 V		2.7 V		2.7 V	2.7 V	5.5 V	E1		20	µA
			46	"															"	1A		"	
			47	"															"	2A		"	
			48	"															"	3A		"	
			49	"															"	4A		"	
			50	"															"	5A		"	
			51	"															"	6A		"	
			52	"															"	E2		"	
	I_{IH2}		53	5.5 V	5.5 V		5.5 V		5.5 V				5.5 V							E1		0.1	mA
			54	"																1A		"	
			55	"																2A		"	
			56	"																3A		"	
			57	"																4A		"	

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}				
1 T_C = 25°C	I_{IH2}	3010	58								GND				5.5 V		5.5 V		5.5 V	5A		0.1	µA
	↓	"	59								"							5.5 V	"	6A		"	"
		"	60								"									$\overline{E2}$			
	I_{IL1}	3009	61	0.4 V	0.4 V		0.4 V		0.4 V		"								"	1A	1/	1/	µA
	↓	"	62	"							"								"	2A	"	"	"
		"	63	"							"								"	3A	"	"	"
		"	64								"		0.4 V						"	4A	"	"	"
		"	65								"				0.4 V			0.4 V	"	5A	"	"	"
		"	66								"						0.4 V	0.4 V	"	6A	"	"	"
	I_{IL2}	"	67	0.4 V							"								"	$\overline{E1}$	"	"	"
	↓	"	68								"							0.4 V	"	$\overline{E2}$	"	"	"
	I_{OS}	3011	69	0.7 V	5.5 V	GND	5.5 V	GND	5.5 V	GND	"								"	1Y	2/	2/	mA
	↓	"	70	"							"								"	2Y	"	"	"
		"	71	"							"								"	3Y	"	"	"
		"	72								"		5.5 V						"	4Y	"	"	"
		"	73								"	GND		GND	6.5 V			0.7 V	"	5Y	"	"	"
		"	74								"					GND	5.5 V	0.7 V	"	6Y	"	"	"
	I_{CC}	3005	75	4.5 V	GND		GND		GND		"		GND		GND		GND	4.5 V	"	V_{CC}		24	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = 125°C.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH}	3003 (Fig. 4)	76	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	16	ns
	↓	"	77	"							"							GND	"	2A - 2Y	"	"	"
		"	78	"							"								"	3A - 3Y	"	"	"
		"	79								"								"	4A - 4Y	"	"	"
		"	80								"								"	5A - 5Y	"	"	"
		"	81								"								"	6A - 6Y	"	"	"
	t_{PHL}	"	82	GND	3/	OUT	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	GND	"	1A - 1Y	"	22	"
	↓	"	83	"							"							GND	"	2A - 2Y	"	"	"
		"	84	"							"								"	3A - 3Y	"	"	"
		"	85								"								"	4A - 4Y	"	"	"
		"	86								"								"	5A - 5Y	"	"	"
		"	87								"								"	6A - 6Y	"	"	"
	t_{PZH}	"	88	4/	4.5 V	OUT					"								"	$\overline{E1}$ - 1Y	"	35	"
	t_{PZL}	"	89	4/	GND	"					"								"	$\overline{E1}$ - 1Y	"	40	"
	t_{PHZ}	"	90	4/	4.5 V	"					"								"	$\overline{E1}$ - 1Y	"	30	"
	t_{PLZ}	"	91	4/	GND	"					"								"	$\overline{E1}$ - 1Y	"	35	"
10 T_C = 125°C	t_{PLH}	"	92	GND	3/	"	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	GND	"	1A - 1Y	"	21	"
	↓	"	93	"							"							GND	"	2A - 2Y	"	"	"
		"	94	"							"								"	3A - 3Y	"	"	"
		"	95								"								"	4A - 4Y	"	"	"
		"	96								"								"	5A - 5Y	"	"	"
		"	97								"								"	6A - 6Y	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}				
10 T_C = 125°C	t_{PHL} ↓	3003 (Fig. 4)	98	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	29	ns
		"	99	"							"							GND	"	2A - 2Y	"	"	"
		"	100	"							"								"	3A - 3Y	"	"	"
		"	101								"								"	4A - 4Y	"	"	"
		"	102								"								"	5A - 5Y	"	"	"
		"	103								"								"	6A - 6Y	"	"	"
	t_{PZH}	"	104	4/	4.5 V	OUT					"								"	$\overline{E1}$ - 1Y	"	45	"
	t_{PZL}	"	105	4/	GND	"					"								"	$\overline{E1}$ - 1Y	"	52	"
	t_{PHZ}	"	106	4/	4.5 V	"					"								"	$\overline{E1}$ - 1Y	"	39	"
	t_{PLZ}	"	107	4/	GND	"					"								"	$\overline{E1}$ - 1Y	"	45	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ I_{IL} limits (μA) min/max values for circuits are shown below.
2/ I_{OS} limits (mA) min/max values for circuits are shown below.

Parameter	Terminals	Circuits A	B	C	D	E	F
I_{IL1}	1A,2A,3A,4A,5A,6A	-150/-380	-30/-300	-160/-400	-160/-400	-160/-400	-160/-400
I_{IL2}	$\overline{E1}$, $\overline{E2}$	-1.0/-150	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{OS}	1Y,2Y,3Y,4Y,5Y,6Y	-40/-225	-30/-130	-30/-130	-15/-100	-30/-130	-15/-100

3/ Pulse from input pulse generator.
4/ Pulse from disable generator.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	0.7 V	0.7 V	-1 mA					GND								4.5 V	1Y	2.4		V dc
		"	2	"			0.7 V	-1 mA			"								"	2Y	"		"
		"	3	"					0.7 V	-1 mA	"								"	3Y	"		"
		"	4	"							"	-1 mA	0.7 V						"	4Y	"		"
		"	5	"							"			-1 mA	0.7 V			0.7 V	"	5Y	"		"
		"	6	"							"					-1 mA	0.7 V	0.7 V	"	6Y	"		"
	V_{OL}	3007	7	0.7 V	2.0 V	12 mA					"								"	1Y		0.4	"
		"	8	"			2.0 V	12 mA			"								"	2Y		"	"
		"	9	"					2.0 V	12 mA	"								"	3Y		"	"
		"	10								"	12 mA	2.0 V						"	4Y		"	"
		"	11								"			12 mA	2.0 V			0.7 V	"	5Y		"	"
		"	12								"					12 mA	2.0 V	0.7 V	"	6Y		"	"
	I_{O1}	"	13	2.0 V	2.0 V	2.4 V					"								5.5 V	1Y		20	µA
		"	14	"	0.7 V	2.4 V					"								"	1Y		"	"
		"	15	"			2.0 V	2.4 V			"								"	2Y		"	"
		"	16	"			0.7 V	2.4 V			"								"	2Y		"	"
		"	17	"					2.0 V	2.4 V	"								"	3Y		"	"
		"	18	"					0.7 V	2.4 V	"								"	3Y		"	"
		"	19	"							"	2.4 V	2.0 V						"	4Y		"	"
		"	20	"							"	2.4 V	0.7 V						"	4Y		"	"
		"	21	"							"			2.4 V	2.0 V			2.0 V	"	5Y		"	"
		"	22	"							"			2.4 V	0.7 V			"	"	5Y		"	"
		"	23	"							"					2.4 V	2.0 V	"	"	6Y		"	"
		"	24	"							"					2.4 V	0.7 V	"	"	6Y		"	"
	I_{O2}	"	25	2.0 V	0.7 V	0.4 V					"								5.5 V	1Y		-20	µA
		"	26	"	2.0 V	0.4 V					"								"	1Y		"	"
		"	27	"			0.7 V	0.4 V			"								"	2Y		"	"
		"	28	"			2.0 V	0.4 V			"								"	2Y		"	"
		"	29	"					0.7 V	0.4 V	"								"	3Y		"	"
		"	30	"					2.0 V	0.4 V	"								"	3Y		"	"
		"	31	"							"	0.4 V	0.7 V						"	4Y		"	"
		"	32	"							"	0.4 V	2.0 V						"	4Y		"	"
		"	33	"							"			0.4 V	0.7 V			2.0 V	"	5Y		"	"
		"	34	"							"			0.4 V	2.0 V			"	"	5Y		"	"
		"	35	"							"					0.4 V	0.7 V	"	"	6Y		"	"
		"	36	"							"					0.4 V	2.0 V	"	"	6Y		"	"
	V_{IC}	"	37		-18 mA						"								4.5 V	1A		-1.5	V dc
		"	38				-18 mA				"								"	2A		"	"
		"	39						-18 mA		"								"	3A		"	"
		"	40								"		-18 mA						"	4A		"	"
		"	41								"				-18 mA				"	5A		"	"
		"	42								"						-18 mA		"	6A		"	"
		"	43	-18 mA							"								"	$\overline{E1}$		"	"
		"	44								"							-18 mA	"	$\overline{E2}$		"	"
	I_{IH1}	3010	45	2.7 V							"								5.5 V	$\overline{E1}$		20	µA
		"	46		2.7 V						"								"	1A		"	"
		"	47				2.7 V				"								"	2A		"	"
		"	48						2.7 V		"								"	3A		"	"
		"	49								"		2.7 V						"	4A		"	"
		"	50								"				2.7 V				"	5A		"	"
		"	51								"						2.7 V		"	6A		"	"
		"	52								"							2.7 V	"	$\overline{E2}$		"	"
	I_{IH2}	"	53	5.5 V							"								"	$\overline{E1}$		0.1	mA
		"	54		5.5 V						"								"	1A		"	"
		"	55				5.5 V				"								"	2A		"	"
		"	56						5.5 V		"								"	3A		"	"
		"	57								"		5.5 V						"	4A		"	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}				
1 T_C = 25°C	I_{IH2}	3010	58								GND				5.5 V		5.5 V		5.5 V	5A		0.1	mA
	↓	"	59								"							5.5 V	"	6A		"	"
		"	60								"									$\overline{E}2$			
	I_{IL1}	3009	61	0.4 V	0.4 V						"								"	1A	1/	1/	μA
		"	62	"			0.4 V				"								"	2A	"	"	"
		"	63	"					0.4 V		"								"	3A	"	"	"
		"	64	"							"		0.4 V						"	4A	"	"	"
		"	65								"				0.4 V			0.4 V	"	5A	"	"	"
	↓	"	66								"						0.4 V	0.4 V	"	6A	"	"	"
	I_{IL2}	"	67	0.4 V							"								"	$\overline{E}1$	"	"	"
	↓	"	68								"							0.4 V	"	$\overline{E}2$	"	"	"
	I_{OS}	3011	69	0.7 V	GND	GND					"								"	1Y	2/	2/	mA
		"	70	"			GND	GND	GND	GND	"								"	2Y	"	"	"
		"	71	"							"	GND	GND						"	3Y	"	"	"
		"	72	"							"			GND	GND				"	4Y	"	"	"
		"	73								"					GND		0.7 V	"	5Y	"	"	"
	↓	"	74								"						GND	0.7 V	"	6Y	"	"	"
	I_{CC}	3005	75	4.5 V	GND		GND		GND		"		GND		GND		GND	4.5 V	"	V_{CC}	"	21	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = 125°C.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH}	3003 (Fig. 4)	76	GND	3/	OUT	3/	OUT	3/	OUT	GND	OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	15	ns
	↓	"	77	"							"							GND	"	2A - 2Y	"	"	"
		"	78	"							"								"	3A - 3Y	"	"	"
		"	79								"								"	4A - 4Y	"	"	"
		"	80								"								"	5A - 5Y	"	"	"
	↓	"	81								"								"	6A - 6Y	"	"	"
	t_{PHL}	"	82	GND	3/	OUT	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	GND	"	1A - 1Y	"	18	"
	↓	"	83	"							"							GND	"	2A - 2Y	"	"	"
		"	84	"							"								"	3A - 3Y	"	"	"
		"	85								"								"	4A - 4Y	"	"	"
		"	86								"								"	5A - 5Y	"	"	"
	↓	"	87								"								"	6A - 6Y	"	"	"
	t_{PZH}	"	88	4/	GND	OUT					"								"	$\overline{E}1$ - 1Y	"	35	"
	t_{PZL}	"	89	4/	4.5 V	"					"								"	$\overline{E}1$ - 1Y	"	45	"
	t_{PHZ}	"	90	4/	GND	"					"								"	$\overline{E}1$ - 1Y	"	32	"
	t_{PLZ}	"	91	4/	4.5 V	"					"								"	$\overline{E}1$ - 1Y	"	35	"
10 T_C = 125°C	t_{PLH}	"	92	GND	3/	"	3/	OUT	3/	OUT	"	OUT	3/	OUT	3/	OUT	3/	GND	"	1A - 1Y	"	20	"
	↓	"	93	"							"							GND	"	2A - 2Y	"	"	"
		"	94	"							"								"	3A - 3Y	"	"	"
		"	95								"								"	4A - 4Y	"	"	"
		"	96								"								"	5A - 5Y	"	"	"
	↓	"	97								"								"	6A - 6Y	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be High ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, F Test no.	1 $\overline{E1}$	2 1A	3 1Y	4 2A	5 2Y	6 3A	7 3Y	8 GND	9 4Y	10 4A	11 5Y	12 5A	13 6Y	14 6A	15 $\overline{E2}$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = 125°C	t_{PHL}	3003 (Fig. 4)	98	GND	3/	OUT	3/	OUT	3/	OUT		OUT	3/	OUT	3/	OUT	3/	GND	5.0 V	1A - 1Y	2	23	ns
			99	"														GND	"	3A - 2Y	"	"	"
			100	"															"	3A - 3Y	"	"	"
			101																"	4A - 4Y	"	"	"
			102																"	5A - 5Y	"	"	"
			103																"	6A - 6Y	"	"	"
	t_{PZH}	"	104	4/	GND	OUT					GND								"	$\overline{E1}$ - 1Y	"	45	"
	t_{PZL}	"	105	"	4.5 V	"					"								"	$\overline{E1}$ - 1Y	"	58	"
	t_{PHZ}	"	106	"	GND	"					"								"	$\overline{E1}$ - 1Y	"	42	"
	t_{PLZ}	"	107	"	4.5 V	"					"								"	$\overline{E1}$ - 1Y	"	45	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ I_{IL} limits (μA) min/max values for circuits are shown below.
2/ I_{OS} limits (mA) min/max values for circuits are shown below.

Parameter	Terminals	Circuits A	B	C	D	E	F
I_{IL1}	1A,2A,3A,4A,5A,6A	-150/-380	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{IL2}	$\overline{E1}$, $\overline{E2}$	-1.0/-150	-30/-300	-160/-400	-160/-400	-160/-400	-150/-380
I_{OS}	1Y,2Y,3Y,4Y,5Y,6Y	-40/-225	-30/-130	-30/-130	-15/-100	-30/-130	-15/-100

3/ Pulse from input pulse generator.
4/ Pulse from disable generator.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.4.5 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as specified herein.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
V_{IN}	Voltage level at an input terminal
V_{IC}	Input clamp voltage
I_{IN}	Current flowing into an input terminal
t_{PHZ}	Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high-impedance (off) state

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t_{PLZ} - - - - - - - - - - - -	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH} - - - - - - - - - - - -	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.
t_{PZL} - - - - - - - - - - - -	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.
T_C - - - - - - - - - - - - - -	Case temperature

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS365
02	54LS366
03	54LS367
04	54LS368

6.7 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table IV.

TABLE IV. Manufacturers' designations.

Military device type	Company						Generic-industry type
	Texas Instruments	Signetics Corp.	Motorola Inc.	Raytheon Company	Fairchild Semiconductor	National Semiconductor	
01	A	B	C	D	E	F	54LS365
02	A	B	C	D	E	F	54LS366
03	A	B	C	D	E	F	54LS367
04	A	B	C	D	E	F	54LS368

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-38510/322B MICROCIRCUITS, DIGITAL, TTL, LOW POWER SCHOTTKY, HEX BUS DRIVERS WITH 3-STATE OUTPUTS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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